



GE VERNOVA

ADMS

16.25.1

Substation Editor Installation Guide

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Change History

Date	Change Description
Jan 26, 2023	ADMS 3.16: • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
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1. Installing the Substation Editor

The Substation Editor is normally installed as part of the ADMS client tool kit. However, it is a stand-alone application, and it can be installed separately if required.

Once the ADMS client tool kit has been installed, the Substation Editor can be started from the Start menu under Programs > Eterra > Distribution > Tools > Modeling Tools > Substation Editor.

1.1. Internationalization

Writing software that accommodates differences in language, culture, and hardware is called "internationalization". The goal of internationalization is to present users with a consistent look, feel, and functionality across different language editions of a product.

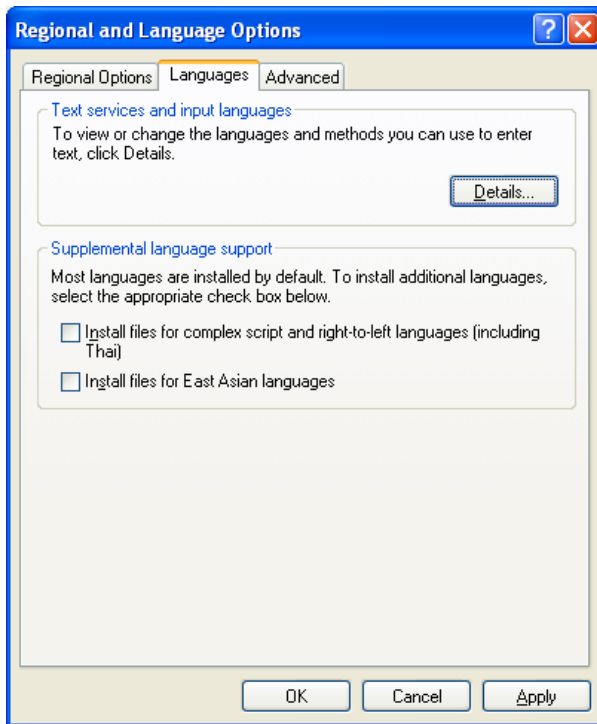
When working with East Asian languages (including Chinese, Japanese, and Korean), internationalization support must be installed manually. For most other languages, Microsoft Windows installs the internationalization files automatically.

Note that a minimum of 230 MB of disk space is required to install additional language files.

Microsoft Windows Vista natively supports East Asian characters. Therefore, for a computer running Windows Vista, no user actions are needed.

To enable East Asian internationalization support in Microsoft Windows Server 2003, perform the following steps:

1. Open the Windows Control Panel.
2. Click Regional and Language Options.
3. Open the Languages tab.



4. Select the "Install files for East Asian languages" check box.
5. Click the Apply button.

You will be prompted to insert the Windows CD-ROM, or to point to a network location where the files are located.

6. After the internationalization files are installed, restart the computer.

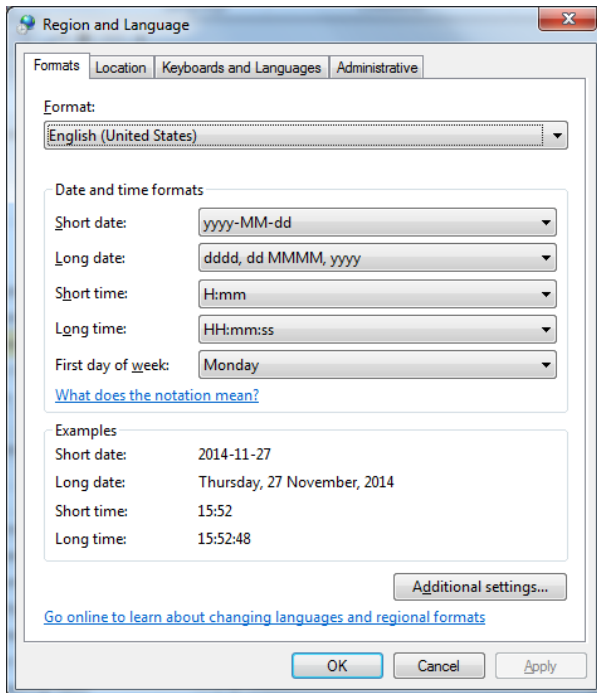
1.2. Setting a Language for Non-Unicode Programs in Windows

The Windows locale determines the default character set (letters, symbols, and numbers) and font that are used to enter and display information in programs that do not use Unicode. This allows non-Unicode programs to run using the specified language.

To specify a language for non-Unicode programs, perform the following steps:

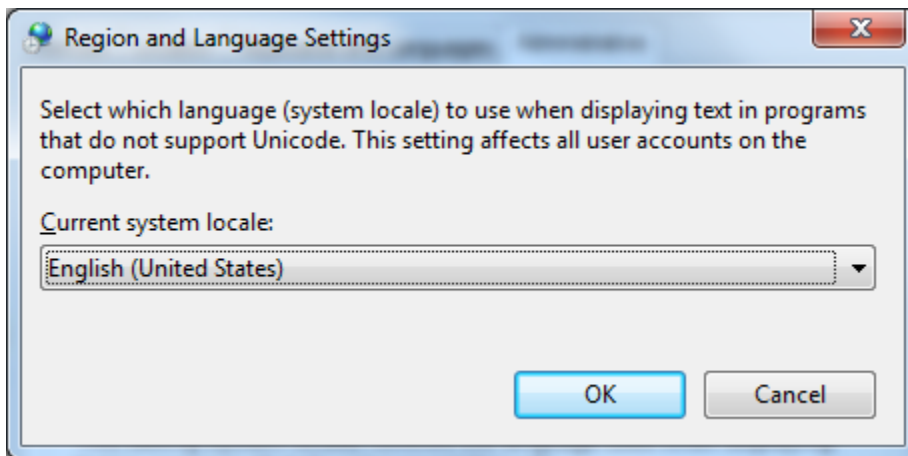
1. Click Start > Control Panel.
2. Click the Region and Language option.

The Region and Language dialog opens.



3. On the Administrative tab, click the Change System Locale button.

The Region and Language Settings dialog opens.



4. Select the language/locale from the drop-down menu.
5. Click OK.

Clicking the OK button closes the Regional and Language Settings window and navigates back to the parent window (Region and Language dialog box).

6. In the Regional and Language window, click OK.

2. Configuring the Substation Editor

For most operations, the default configuration of the Substation Editor is suitable. Configuration changes should only be made at the administrative level. There should be no need to change the configuration on a user-to-user basis.

The operating parameters of the Substation Editor can be configured from the Editor Configuration pop-up. This is selected from "Tools" on the menu bar. The configuration is split into ten pages, which are selected from the tabs at the bottom of the window:

- Associations
- General
- External Station Configurations
- Topology
- Linker
- Global Internals
- Template Names
- Poly Placements
- SCADA Configurations
- Colors & Halos

The actual configuration is saved as an XML file in the same location that it was imported from. The configuration file to use is normally the one that is specified in the Windows shortcut used to start the Substation Editor.

Once the configuration changes have been made, they need to be applied to the Editor by clicking the Apply button at the bottom of the window.

Clicking the Cancel button closes the configuration pop-up without applying any changes.

Clicking the OK button applies the changes and closes the window.

2.1. Associations

Associations are used to establish the relationship between the elements that are used on the display and the DNOM objects that are instantiated as part of the display building process.

The screenshot displays the 'Substation Editor Configuration' dialog box, specifically the 'Associations' tab. The interface is organized into several sections:

- Top Section:** Includes a radio button for 'Associated Element' (selected), a dropdown for 'Add Association', a checkbox for 'Mismatch' with a 'Copy' button, a 'DNOM Type' dropdown, and an 'Assoc. Reference Picture' dropdown.
- Connection and Label Section:** Contains input fields for 'Connection Pt 1', 'Connection Pt 2', and 'Connection Pt 3'. It also has checkboxes for 'Reference Only', 'Three Winding Transformer', 'Create DNOM Symbol' (checked), and 'Create Label'. A 'FG Layer' dropdown and buttons for 'Delete Mismatch Assoc.' and 'Clear Assoc.' are also present.
- DNOM Symbol and Label Section:** Features input fields for 'Offset' and 'Rotation' for both the 'DNOM Symbol' and 'DNOM Label'.
- Measurement Info Section:** Includes dropdowns for 'MEAS Type', 'MEAS Group', 'Default Assoc.', and 'Default CAM', along with a 'Create Meas' checkbox.
- Bottom Section:** Contains a 'Composite Key' input field, 'SCADA DataSource' and 'SCADA Model Name' dropdowns, a 'Refresh' button, and a 'Division' input field. A 'Controls' section with a dropdown is also visible.
- Table:** A large table on the right side of the dialog, with columns for 'Name' and 'Value', intended for listing and managing associations.
- Footer:** A tabbed interface at the bottom shows 'Associations' as the active tab, with other tabs like 'General', 'External Station Configurations', 'Topology', 'Linker', 'Global Internals', 'Template Names', 'Poly Placements', 'SCADA Configurations', and 'Colors & Halos'. 'OK', 'Apply', and 'Cancel' buttons are at the bottom right.

Figure 1. Substation Editor Configuration – Associations

The Associations tab provides a table of modeling parameters that can be set for each display element/object. For more details about the parameters that may be used, refer to the index.

It is important to note that, while each of the display elements can be associated with each of the DNOM object types, many of these combinations are not relevant and, therefore, will have no parameters defined.

The Associated Element combo box shows only elements that have been associated with a DNOM object type or that have some configuration option set.

To add new associations, select the Add Association radio button, then select the name of the element to add associations for in the Add Association combo box. Proceed to set the properties needed for the association.

In many cases, an association already exists for an element that is identical or very close to another association. For example, an association exists for a vertical switch element, but not for a horizontal switch element. In this case, select the Add Association radio button, then select the name of the element to add associations for in the Add Association combo box. To activate the Copy functionality, select the left check box below the Add Association radio button. In the combo box next to the Copy button, select the existing association to copy from. After selecting the association

to copy from, click the Copy button.

A Mismatch check box is located to the left of the Copy button. This can be used with a substation configuration file, such as the one delivered as an example in the installation kit, as a starting point for making customer-specific configurations. In this case, there are existing associations that do not match the element names in the customer's element set. In order to associate a previously configured association with the customer's element set, select the Mismatch check box, select the "Mismatched" association in the combo box, and then click the Copy button.

The Assoc. Reference Picture field is provided to define the picture that will be displayed when a reference-only object is created by mapping an internal object.

In cases where the association is no longer valid (i.e., the object no longer exists in the element set), the Delete Mismatch Assoc. button will be enabled to allow the association to be cleared. The network object can then be associated with a new object in the element set, if required.

In the case where the association between a display element and a network object needs to be changed, the Clear Assoc. button can be used to clear an existing association. Neither of the objects is affected; only the association is removed.

Where changes have been made to the associations as part of the Substation Editor configuration, it may be necessary to update existing or partially built displays. This is done in the normal display building process by selecting the device on the display and clicking the Reapply Associations for Placement button on the main toolbar. Each device can be updated separately, or the Reapply Associations for View button can be used to update the associations for all placements in the view. Each view must be updated individually.

Note that the "Reapply Associations" logic is purposely conservative. It will only "add" property values not previously configured. If a property value has already been set, the "Reapply Association" logic will not change the existing value. This is done to prevent an intentionally set value from being accidentally overwritten.

2.2. General

The General tab provides a means of selecting the location of the display and menu element sets and the auto-save interval for the editing session.

Substation Editor Configuration

Element Set: [%DMS_ROOT%\Tools\workspace\SubstationEditor\Fantasyisland_internals_eset.dll]

Element Set for Menus: [%DMS_ROOT%\Tools\workspace\SubstationEditor\Fantasyisland_internals_eset.dll]

Default Display Properties

Application(s): SCADA.RECON

Background Color: Black

Calldown Command:

Callup Command: SET KEY222= %SUBSTN%

Permission Test:

Title: %DIS% %DISAPP% %HOST% [%VP%] %REF%

Menu

Label	Element Set	Name	Application
Related Displays	FANTASYISLAND_INTERNA...	SCADA_DISPLAYS	
Scada Onelines	FANTASYISLAND_INTERNA...	SCADA_ONELINES	
Tagging/Notes	FANTASYISLAND_INTERNA...	SCADA_TAGGING_NOTES	

☒ No Default Menu Bar Menus

Import Menu Element Set

e-terraBrowser DDL Compatibility Version: WebFG 3.6.0

Safety Doc Layer

☐ Export Safety Doc Layer

Driving App: SAFETY Driving Db: SAFETY Link Field: D\$DOC_DOC Element Set: FANTASYISLAND_INTERNALS

Display Notes Layer

☐ Export Display Notes Layer

Driving App: TAGGING Driving Db: TAGGING Link Field: F\$NOTE_NOTE Element Set: AUTOGEN

Auto Generation Configuration

Feeder Picture: FDRHEAD_L

Interface Picture: INTERFACE_V

Connection Picture: CONNECTION_D

AutoSave Interval: 0 Minutes (0 to disable)

Encoding for non-Unicode files: Western European (Windows) - Windows-1252

Log File Properties

Log File Path: [%DMS_ROOT%\log]

Number of Saved Log Files: 2 Log Level: Info

Model Manager Properties

Autoprocess flag: ☐ true ☒ false

Associations | General | External Station Configurations | Topology | Linker | Global Internals | Template Names | Poly Placements | SCADA Configurations | Colors & Halos

OK Apply Cancel

Figure 2. Substation Editor Configuration – General

Much of the information in this section is used to configure the Browser displays that are exported from the Substation Editor. For more information about the functions and configuration of Browser displays, refer to the *Display Builder User Guide*.

- **Element Sets:** These fields are used to point to the location of the Browser element sets that are to be used in the display generation process.
- **Default Display Properties:** These fields are used to define properties of the Browser display that is exported from the Substation Editor.
- **Menu:** These settings are used to specify the menus that will appear on the main Browser menu bar of the exported display.
- **Safety Doc Layer:** These fields are used to specify whether the Safety Documents Layer should be added to the exported display.

The Driving App field is used to specify the name of the Habitat Safety application, the Driving Db field specifies the name of the Habitat Safety database, and Link Field specifies the database link field for Safety Documents.

- **Log File Path:** This path is used to specify where the trace log file is saved.

- **Model Manager Properties:** This option is used to specify the value of the <AutoProcess> element when creating the ISSG file.

If changes have been made to the element set within the Browser display builder, it is necessary to re-load the element set into the Substation Editor. This is done using the Reload Element Set button located on this page.

Automatic periodic saving of Substation Editor files can be configured by setting a value (minutes) in the AutoSave Intervals. This will cause the Substation Editor to automatically save the current file with the file name Xxxxxx_AutoSave_subeditor.xml

The Auto Generation Configuration panel is used to specify the picture/symbol to be used on the Substation Internals display for the feeder head, interface, and connection nodes. The pictures are selected by using the pull-down menus for each picture type.

Encoding for non-Unicode files allows the selection of the type of character encoding that is to be used in the cases where a non-Unicode character set has been defined for the display.

2.3. External Station Configurations

The External Station Config tab is used to define the mapping between DNOM objects and the pictures used to display reference-only objects.

This tab is used when a substation external model file is imported into the Substation Editor. DNOM objects are graphically selected to create reference-only objects on a display.

The list of DNOM object types is defined by entering the folder for the mapper table file. For each DNOM object type, an associated picture can be specified by selecting it from the pull-down menu.

On the Imported Station Externals display, the font for the display of labels can be set by clicking the Set Font button.

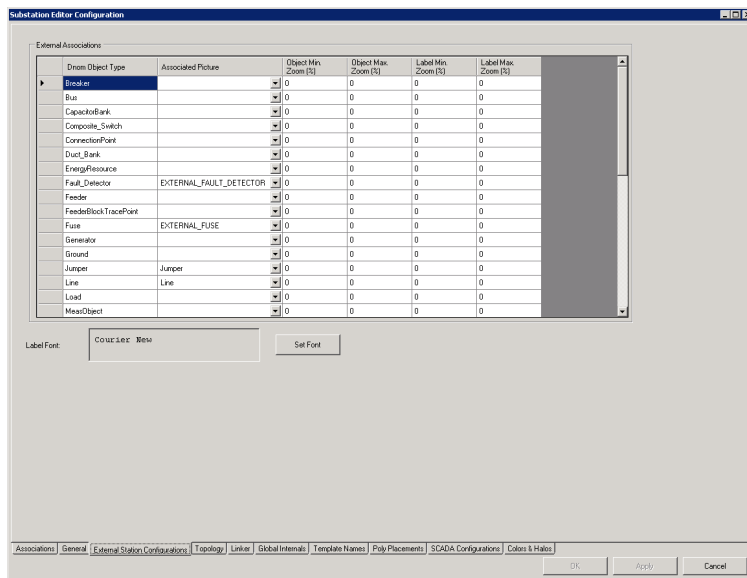


Figure 3. Substation Editor Configuration – External Configurations

2.4. Topology

The Topology Associations group enables the association between some display elements and their characteristics.

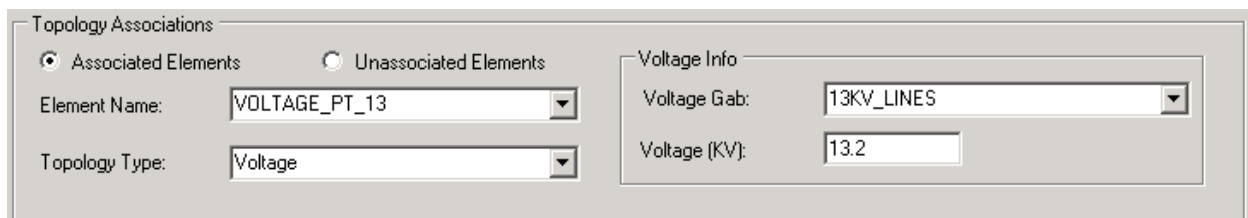


Figure 4. Topology Associations for Voltage Elements

The above image shows the characteristics available for assigning to a picture or symbol. When "Associated Elements" is selected, the drop-down list of Element Names shows the elements for which some or all characteristics have been set. When "Unassociated Elements" is selected, all other elements are made available to be defined as a Voltage or Node_Specifier.

Element Name: The drop-down list of available display elements. **Topology Type:** Voltage or Node_Specifier.

For Voltage Elements only:

- **Voltage Gab:** The FG GAB used to set the color and other graphic characteristics of the line or jumper to which it is attached.
- **Voltage (KV):** The voltage assigned to the device to whose placement the Voltage Gab or the FG element (Element Name) is attached.

Specifying an FG Element as a Node_Specifier, as in the below image, makes the Specific Node ID

property available for the modeler to set the ID of the node to which the placement is connected, rather than having Substation Editor assign it. This is needed, for example, to specify the nodes of a switch that connects two substations (as in a substation-fed-by-substation situation).

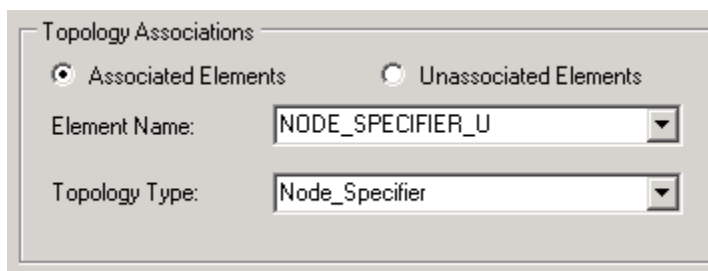


Figure 5. Topology Associations for Node Specifiers

2.5. Linker

The Linker tab allows the definition of the Habitat hosts that are to be used to help form composite key links to the displays (for example, the NETMOM or SCADAMOM databases).

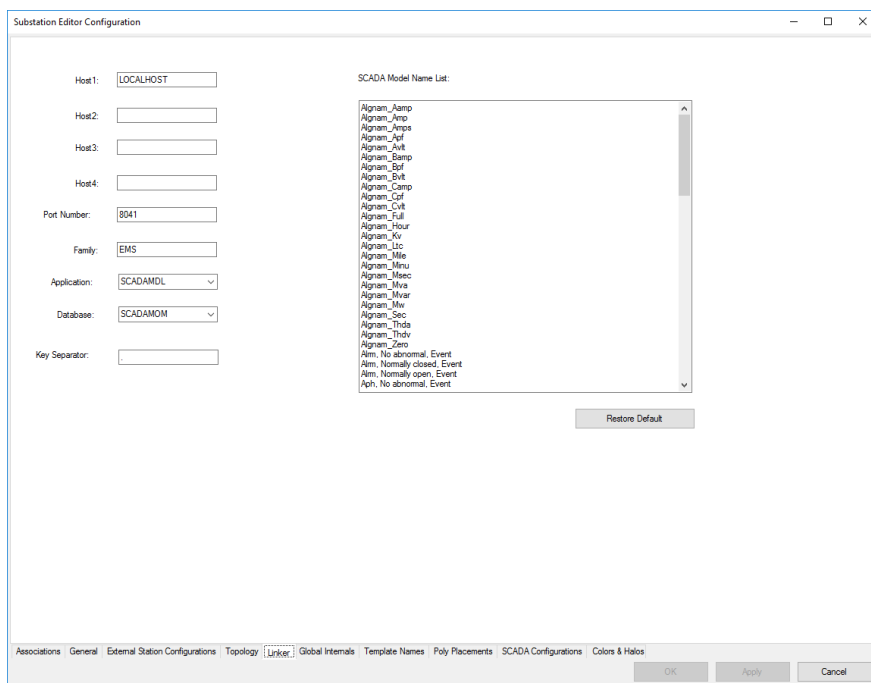


Figure 6. Substation Editor Configuration – Linker

2.6. Global Internals

Global internals are used to define the parameters for the device types found in the substation internals model. This is similar to the global file that is used to define the device type parameters for the standard DNOM models (externals).

The global internals file is used by the converter when creating a station model file (.mod) that includes station internals. The exported file, `global_internals.xml`, must be placed in the converter input folder.

The basic model types are defined in the model object type list. Instances of these object types can be created by clicking the Add New Model button. The new device type is added to the globals list, and the properties are updated as required. [Global Internal Model Object Type Properties](#) lists and explains the properties of the model types and the corresponding field names in `global_internals.xml` and `global.csv`.

Note

The global internals file is exported each time the Configuration, OK, or Apply buttons are clicked. In this way, the substation internals model file and the global file definitions are always consistent.

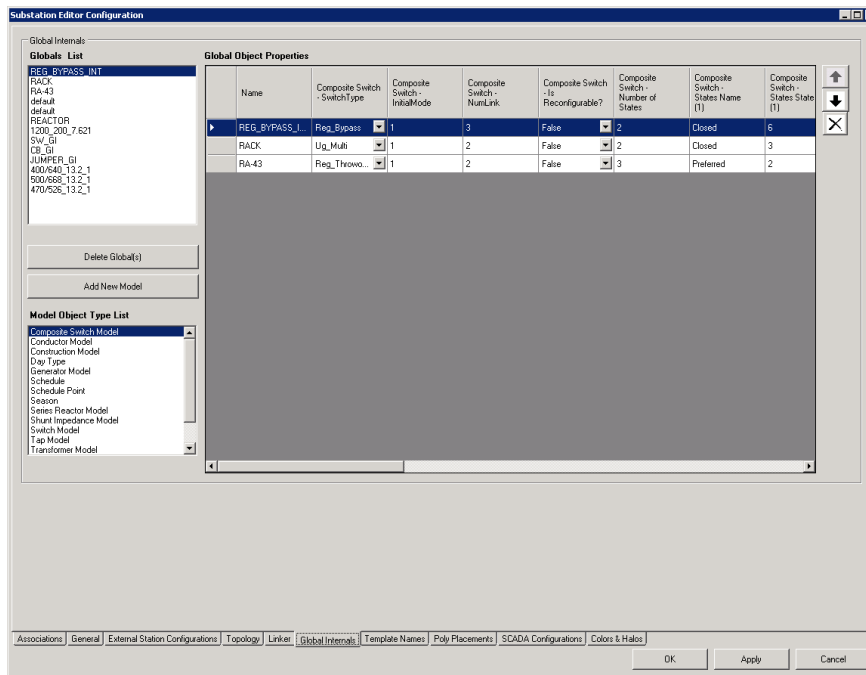


Figure 7. Substation Editor Configuration – Global Internals

2.7. Template Names

This configuration page allows template names to be defined along with the value that will be applied to the name. A new template name can be added to the list by entering it on the bottom line of the list.

The intent of the template names configuration is to be able to configure templates that will help set property values common to most if not all stations modeled.

In the example below, `%DNOMSTATON%` and `%SUBSTN%` represent the long and short name of a

substation. %AREA% represents the permission area assigned to the substation. Each time a new substation is modeled, the user fills in these values once, before stating to model substation equipment. Each time substation equipment is added to the substation, these common template values are filled in wherever needed, without the user actually having to fill them in each time.

Note that the template names configuration allows a template value to be filled in. Normally, the value field is left blank. Only when a value is always going to be the same should a value be filled in.

Substation Editor Configuration		
	Template Name	Template Value
▶	%DNOMSTATION%	<StationName>
	%SUBSTN%	
	%DIVISION%	
*		

Figure 8. Substation Editor Configuration – Template Names

When changes are made to the template names within the Substation Editor configuration, it may be necessary to apply the changes to existing or partially built displays. This is done in the normal display building process by clicking the Reapply Template Values button on the toolbar. All instances of the template values within the display are automatically updated.

2.8. Poly Placements

Default configurations for jumpers, lines, and buses can be entered on this page. For example, the switch model that is to be applied for jumpers is defined here. This avoids having to specify a switch model every time a jumper is placed.

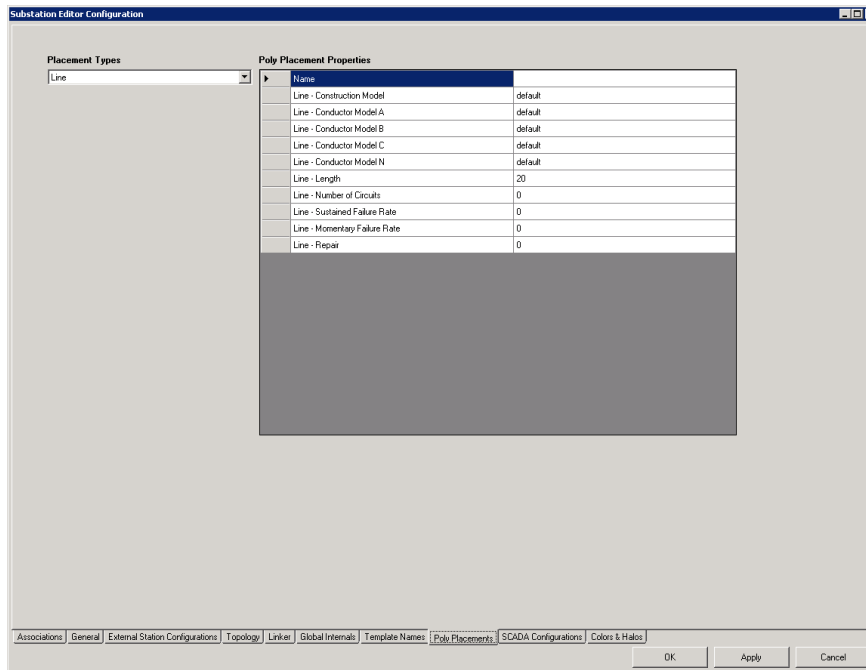


Figure 9. Substation Editor Configuration – Poly Placements

2.9. SCADA Configurations

The SCADA configuration allows the settings of SCADA controls such as Trip, Close, etc., to be configured. The control values are used when exporting Control or Habitat import files, in order to initialize the Scada database as best as possible.

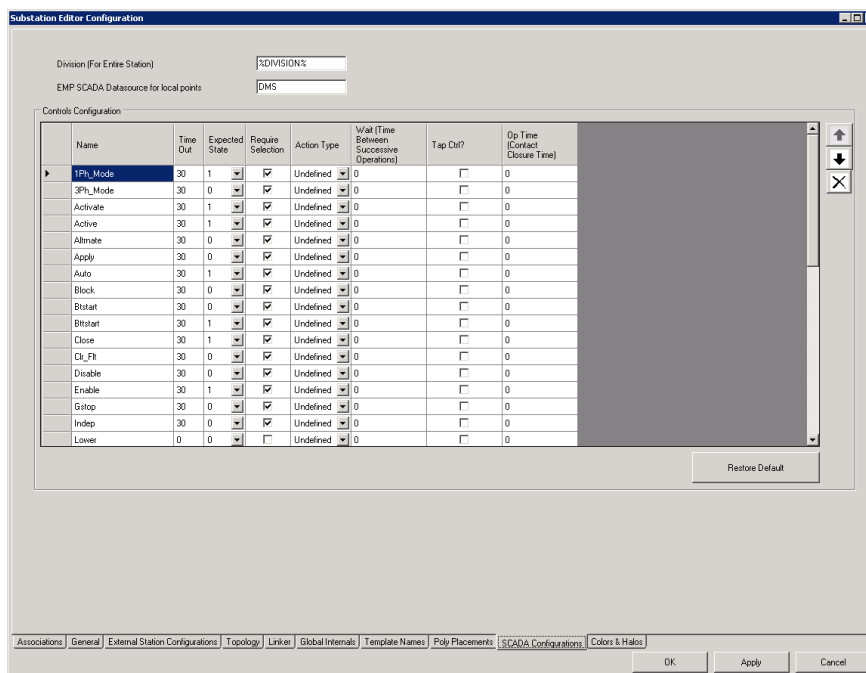


Figure 10. Substation Editor Configuration – SCADA Configurations

2.10. Colors and Halos

2.10.1. About Colors and Halos

Feeder colors, voltage colors, and halos are all configurable in Substation Editor so that SCADA oneline displays can be built with the same coloring conventions as in the ADMS Station Internals views. Analog measurements, which return ADMS line information to SCADA, are modeled for the same purpose.

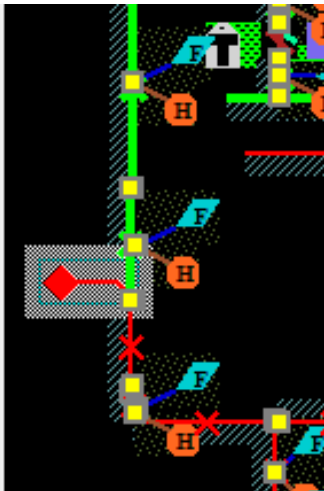
2.10.1.1. Feeder Colors in Normal State

In normal state, a line outside the station fence is colored according to the viewer configuration color assignment for the line's given Feeder Color Number.

In order to have the lines and jumpers of the feeder heads inside the substation match the colors of their respective feeder lines outside the substation, you must coordinate the color specifications as explained in the "Feeder Colors" section in the *ADMS Modeling Reference Guide*.

For example, in the Fantasy Island RAVEN station, Feeder 8861 is given the color "red" (RGB=255,0,0) by attaching the Shared Gab named FEEDERCOLOR1 to its FDRHEAD element, as shown in the below image:

Association Type	Feeder
Attachment - Cam	<None>
Attachment - Key Set	<None>
Attachment - Menu	<None>
Attachment - Shared Gab	FEEDERCOLOR1
Composite Key	
Division (display grouping)	
DNOM X	1440
DNOM Y	-10080
Element Name	FDRHEAD_L



This matches the color assigned as Feeder Color 1 in the viewer configuration, so the internal feeder head color matches the external feeder line color.

2.10.2. Configuring Halos

The Colors & Halos configuration tab allows you to define some of the default colors that are used in display definitions, and to configure the properties of energization halos. You can configure the halos that surround display placements on this tab.

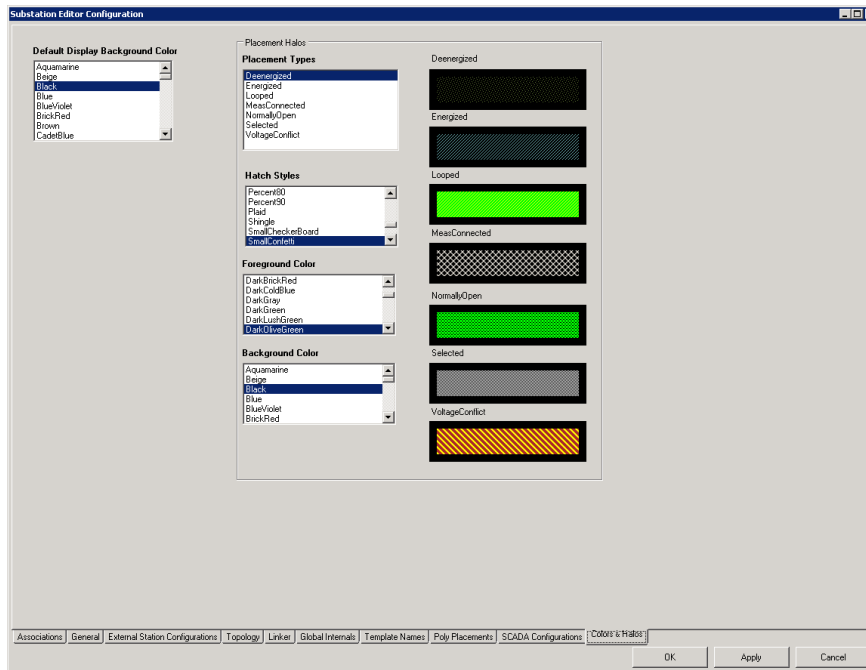


Figure 11. Substation Editor Configuration – Colors & Halo Configuration

The default display background color is defined by selecting one of the colors in the Default Display Background Color list box. The currently defined energization states are each associated with a set of drawing attributes that define a graphic combination, which is drawn behind a placement depending on the placement object's energization state.

The Placement Types list box lists the different energization state types. You can define a hatch style, foreground color, and background color for each type.

3. Command-Line Operations

The normal startup command for the Substation Editor, as used for the Windows shortcut, is shown below:

```
D:\Eterra\Distribution\Tools\bin\SubstationEditor.exe -ConfigFile  
"D:\Eterra\Distribution\Tools\config\SubstationEditorConfiguration.xml"
```

The Substation Editor can also be run directly from the Windows command line for scripted or batch mode operations.

The syntax and command-line arguments for SubstationEditor.exe are as follows:

- **CONFIGFILE:** The configuration file that the Substation Editor opens on startup. This must be specified for all operations.
- **EXPORTFILE** xxxx_subeditor.xml: The name of the Substation Editor file to be used for the export process. The files to be exported are determined by the -EXPORTTYPE argument.
- **EXPORTDIR** XXXXX: A directory containing one or more *_subeditor.xml files to be used for the export process. The files to be exported are determined by the -EXPORTTYPE argument. This argument will override (but not overwrite) the path in the preferences file.
- **MMAUTOPROCESSFLAG:** Specify either "true" or "false" to use the <AutoProcess> element in the ISSG file. The configuration value is used if it is not defined by the command-line parameter.
- **EXPORTTYPE** XXXX: The type of export to perform. The arguments can be:
 - ALL: Export the station internals and placements, the Browser display, and the Control and the Platform databases.
 - SMP: Export an Control database.
 - DDL: Export the Browser display.
 - INTERNALS: Export the station internals and placements.
 - HDB: Export an Platform database.
 - AFS: Export Automatic Feeder Schematic placements(AFS).

Note

To generate AFS placements, substation .mod files must be created with the Converter tool

- **OUT** XXXX: The directory where the exported file will be saved. If this is not specified, the file will be exported to the same directory as the input file.
- **CULTURE** XXXX: The culture name to be used when the user interface should be presented in a locale other than U.S. English.
- **EMPSCADADATASOURCE** XXXX: The SCADA data source to use for all SCADA data source property values left blank in the substation model. This is used only when the EXPORTTYPE HDB

or ALL is specified.

- **EXTERNALSDIR, EXTDIR:** The directory in which to find station externals files.
- **MULTIFILE ASSOC:** The command to reapply associations for multi-file processing to all the substations in INPUTDIR.
- **MULTIFILE TEMPLATE:** The command to reapply template values for multi-file processing to all the substations in INPUTDIR.
- **MULTIFILE CORRECTMEAS:** The command to force an update of the DNOMSTATION for all measurements. Refer to the *Substation Editor User Guide*, "Correcting Measurement Attachments" for more detail about this functionality.
- **INPUTDIR:** A directory containing one or more *_subeditor.xml files on which a MULTIFILE operation is performed.
- **STATIONNAME:** Used when EXPORTTYPE is set to AFS. The Substation name for which AFS placements will be generated. Use ALL to generate AFS placements for all substations.
- **HELP:** The Help menu.

Below are some examples of using the above batch commands. For brevity, we assume the following definitions of environmental variables:

```
set WORKSPACE=%IDMS_ROOT%\ToolsWorkspace
set SSIEDIR=%WORKSPACE%\SubstationEditor
set CONFIG=%SSIEDIR%\SubstationEditorConfiguration.xml
```

Example:

```
SubstationEditor.exe
-ConfigFile %CONFIG%
-ExportFile %SSIEDIR%\data\STATION_subeditor.xml
-ExportType ALL
> MyExport.log 2>&1
```

Example:

```
SubstationEditor.exe
-ConfigFile %CONFIG% -MULTIFILE ASSOC
-MULTIFILE ASSOC
-INPUTDIR %SSIEDIR%\data
> MultifileAssoc.log 2>&1
```

Example:

```
SubstationEditor.exe
-ConfigFile %CONFIG% -MULTIFILE CORRECTMEAS
-EXTDIR %WORKSPACE%\CONVERTER\input
-INPUTDIR %SSIEDIR%\data
> MultifileCorrectMeas.log 2>&1
```

4. Global Internal Model Object Type Properties

4.1. Composite Switch Model

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Name	Name	NAME_COMPSWMOD	Composite switch model name.
Composite Switch – Switch Type	SwitchType	SWITCHTYPE_COMPSWMOD	Enumerated composite switch model type: 1 = Throwover, 2 = ESCO Throwover, 3 = RAL, 4 = GRAL, 5 = Regulator Bypass, 6 = UG Multi Switch.
Composite Switch – InitialMode	InitialMode	INITIALMODE_COMPSWMOD	Initial operating mode (true = automatic, false = manual).
Composite Switch – NumLink	NumLink	NUMLINK_COMPSWMOD	UI
Composite Switch – Is Reconfigurable	Reconfigurable	RECONFIGURE_COMPSWMOD	Reconfigurable flag for geoschematics (true = reconfigurable).
Composite Switch – Number of States	NumStates	NUMSTATE_COMPSWMOD	UI
States(1): State	CIMDnom.Composite_Switch_state	STATEn_COMPSWMOD	Integer value representing the on/off status of the individual switch links. For a throwover switch in the preferred state, the value of the switch links is closed on the preferred link (1) and open on the alternate link (0). This gives a binary state value of 10, which is equal to 2 (base 10).
States(2): State	CIMDnom.Composite_Switch_state	STATEn_COMPSWMOD	Integer value representing the on/off status of the individual switch links. For a throwover switch in the preferred state, the value of the switch links is closed on the preferred link (1) and open on the alternate link (0). This gives a binary state value of 10, which is equal to 2 (base 10).

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
States(3): State	CIMDnom.Composite_Switch_state	STATEn_COMPSWMOD	Integer value representing the on/off status of the individual switch links. For a throwover switch in the preferred state, the value of the switch links is closed on the preferred link (1) and open on the alternate link (0). This gives a binary state value of 10, which is equal to 2 (base 10).
States(4): State	CIMDnom.Composite_Switch_state	STATEn_COMPSWMOD	Integer value representing the on/off status of the individual switch links. For a throwover switch in the preferred state, the value of the switch links is closed on the preferred link (1) and open on the alternate link (0). This gives a binary state value of 10, which is equal to 2 (base 10).

4.2. Conductor Model

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Name	Name	NAME_CONDMOD	Unique conductor model name. SCC: Conductor model name. AMS: oh_primary_wire.type: 10="Aerial"; 30="Bare"; 20="Hendrix"; 9999="Not Listed"; 40="Tree"; 999="Unknown".
Conductor – Is Dist Cond?	DistCond	DISTCOND_CONDMOD	Flag indicating model represents a distribution conductor (true = distribution, false = transmission).
Conductor – Rphase	Rphase	RPHASE_CONDMOD	Phase conductor resistance (ohms/kft) for a transmission or distribution model.
Conductor – GmRphase	GmRphase	GMRPHASE_CONDMOD	Phase conductor geometric mean radius (feet) if a distribution model is given (blank for a transmission model).
Conductor – Gshunt	Gshunt	GSHUNT_CONDMOD	Shunt conductance (Siemens x 10-06/kft) if a transmission model is given (blank for a distribution model).
Conductor – Bshunt	Bshunt	BSHUNT_CONDMOD	Shunt susceptance (Siemens x 10-06/kft) if a transmission model is given (blank for a distribution model).

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Conductor – Xphase	Xphase	XPHASE_CONDMOD	Phase conductor reactance (ohms/kft) if a transmission model is given (blank for a distribution model).
Conductor – Diameter	Diameter	DIAMETER_CONDMOD	Diameter of the conductor (inches) if a distribution model is given (blank for a transmission model).
Conductor – Rneutral	Rneutral	RNEUTRAL_CONDMOD	Neutral conductor resistance (ohms/kft) for a concentric neutral cable if a distribution model is given (blank for a transmission model).
Conductor – GmRneutral	GmRneutral	GMRNEUT_CONDMOD	Neutral conductor bundle self geometric mean radius (feet) for a concentric neutral cable if a distribution model is given (blank for a transmission model).
Conductor – PhDist	PhDist	PHDIST_CONDMOD	Phase conductor spacing (inches) for a three conductor cable if a distribution model is given (blank for a transmission model).
Conductor – NeutDist	NeutDist	NEUTDIST_CONDMOD	Center to neutral conductor spacing (inches) for a concentric neutral cable (blank for a transmission model).
Conductor – Insulation	Insulation	INSULATION_CONDMOD	Type of cable insulation. The enumeration is as follows: 1 = Cross-Linked Polyethylene (XLPE) 2 = Polyethylene (PE) 3 = Polyvinyl Chloride (PVC) 4 = Ethylene-Propylene Rubber (EPR).
Conductor – ShieldThick	ShieldThick	SHIELDTHICK_CONDMOD	Shield thickness (in inches) for a non-concentric neutral cable if a distribution model is given (blank for a transmission model).
Conductor – Diam. Shield	DIAMShield	DIAMSHIELD_CONDMOD	Outside diameter over the shield (inches) for a non-concentric neutral cable if a distribution model is given (blank for a transmission model).
Conductor – Number of Strands	NumStrands	NUMSTRANDS_CONDMOD	The number of concentric neutral strands for a concentric neutral cable if a distribution model is given (blank for a transmission model).

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Conductor – Diam. Neutral	DiamNeutral	DIAMNEUTRAL_CONDMOD	Diameter (inches) of a concentric neutral strand for a concentric neutral cable if a distribution model is given (blank for a transmission model).
Conductor – Cable Type	CableType	CABLETYPE_CONDMOD	Enumerated cable type. The following enumeration is used: 0 = Bare wire (includes tree wire), 1 = Single tape shield cable, 2 = One three conductor tape shield cable, 3 = Single concentric neutral cable, 4 = One three conductor concentric neutral cable.
Conductor – SustfRate	SustfRate	SUSTFRATE_CONDMOD	Default sustained failure rate (permanent failures/kft-year).
Conductor – MomfRate	MomfRate	MOMFRATE_CONDMOD	Default momentary failure rate (temporary failures/kft-year).
Conductor – MTTR	MTTR	MTTR_CONDMOD	Default mean time to repair (MTTR – hours).
Conductor – AmpDerateSched	AmpDerateSched	AMPDERATESCHED_CONDMOD	Ampacity derating schedule name for cases when there are multiple cables in a duct bank. This schedule is not required.
Conductor – SmallDisp?	SmallDisp	SMALLDISP_CONDMOD	Flag indicating that a small conductor type has been found – used for displaying lines (true = small conductor).
Conductor – TypeDesc	TypeDesc	TYPEDESC_CONDMOD	Conductor type (copper, other).
Conductor – SizeDesc	SizeDesc	SIZEDESC_CONDMOD	Conductor size (568, 1.0, other).
Conductor – CnstType	CnstType	CNSTTYPE_CONDMOD	Enumeration indicating that underground construction is direct buried (0), in conduit (1), or not applicable (2 – overhead construction).
Oplimits(1): Season	CIMDnom.ConductorOpLimit.Season	OPLIMITSEASONn_CONDMOD	The name of an associated season when the limit is applicable. If no season is given, then the limit is applied for the entire year.
Oplimits(1): TempSched	CIMDnom.ConductorOpLimit.TempSched	AMPACITYSCHEDn_CONDMOD	The name of an associated temperature based ampacity schedule.
Oplimits(1): Name	CIMDnom.ConductorOpLimit.Name	OPLIMITNAMEn_CONDMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Oplimits(1): Unit	CIMDnom.ConductorOpLimit.Unit	OPLIMITUNITn_CONDMOD	Enumerated limit units, 1 = amps, 2 = kVA.
Oplimits(1): Limit	CIMDnom.ConductorOpLimit.Limit	OPLIMITn_CONDMOD	The value of the limit (in amps or kVA).
Oplimits(2): Season	CIMDnom.ConductorOpLimit.Season	OPLIMITSEASONn_CONDMOD	The name of an associated season when the limit is applicable. If no season is given, then the limit is applied for the entire year.
Oplimits(2): TempSched	CIMDnom.ConductorOpLimit.TempSched	AMPACITYSCHEDn_CONDMOD	The name of an associated temperature based ampacity schedule.
Oplimits(2): Name	CIMDnom.ConductorOpLimit.Name	OPLIMITNAMEn_CONDMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(2): Unit	CIMDnom.ConductorOpLimit.Unit	OPLIMITUNITn_CONDMOD	Enumerated limit units, 1 = amps, 2 = kVA.
Oplimits(2): Limit	CIMDnom.ConductorOpLimit.Limit	OPLIMITn_CONDMOD	The value of the limit (in amps or kVA).
Oplimits(3): Season	CIMDnom.ConductorOpLimit.Season	OPLIMITSEASONn_CONDMOD	The name of an associated season when the limit is applicable. If no season is given, then the limit is applied for the entire year.
Oplimits(3): TempSched	CIMDnom.ConductorOpLimit.TempSched	AMPACITYSCHEDn_CONDMOD	The name of an associated temperature based ampacity schedule.
Oplimits(3): Name	CIMDnom.ConductorOpLimit.Name	OPLIMITNAMEn_CONDMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(3): Unit	CIMDnom.ConductorOpLimit.Unit	OPLIMITUNITn_CONDMOD	Enumerated limit units, 1 = amps, 2 = kVA.
Oplimits(3): Limit	CIMDnom.ConductorOpLimit.Limit	OPLIMITn_CONDMOD	The value of the limit (in amps or kVA).
Oplimits(4): Season	CIMDnom.ConductorOpLimit.Season	OPLIMITSEASONn_CONDMOD	The name of an associated season when the limit is applicable. If no season is given, then the limit is applied for the entire year.
Oplimits(4): TempSched	CIMDnom.ConductorOpLimit.TempSched	AMPACITYSCHEDn_CONDMOD	The name of an associated temperature based ampacity schedule.
Oplimits(4): Name	CIMDnom.ConductorOpLimit.Name	OPLIMITNAMEn_CONDMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(4): Unit	CIMDnom.ConductorOpLimit.Unit	OPLIMITUNITn_CONDMOD	Enumerated limit units, 1 = amps, 2 = kVA.

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Oplimits(4): Limit	CIMDnom.ConductorOpLimit.Limit	OPLIMITn_CONDMOD	The value of the limit (in amps or kVA).
Oplimits(5): Season	CIMDnom.ConductorOpLimit.Season	OPLIMITSEASONn_CONDMOD	The name of an associated season when the limit is applicable. If no season is given, then the limit is applied for the entire year.
Oplimits(5): TempSched	CIMDnom.ConductorOpLimit.TempSched	AMPACITYSCHEDn_CONDMOD	The name of an associated temperature based ampacity schedule.
Oplimits(5): Name	CIMDnom.ConductorOpLimit.Name	OPLIMITNAMEn_CONDMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(5): Unit	CIMDnom.ConductorOpLimit.Unit	OPLIMITUNITn_CONDMOD	Enumerated limit units, 1 = amps, 2 = kVA.
Oplimits(5): Limit	CIMDnom.ConductorOpLimit.Limit	OPLIMITn_CONDMOD	The value of the limit (in amps or kVA).

4.3. Construction Model

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Name	Name	NAME_CNSTMOD	Framing model name.
Construction – Voltage	Voltage	VOLTAGE_CNSTMOD	Voltage class (15 kV, 25 kV, other).
Construction – OHUG Flag	OHUGFlag	OHUGFLAG_CNSTMOD	Flag indicating overhead vs. underground construction (True=overhead).
Construction – Xa Coordinate	XaCoord	XACoord_CNSTMOD	X coordinate for conductor on position 1(vertical coordinates – feet). Usually, position1 is the leftmost position, which means its X coordinate is smallest among all positions.
Construction – Ya Coordinate	YaCoord	YACoord_CNSTMOD	Y coordinate for conductor on position1 (vertical coordinates – feet).
Construction – Xb Coordinate	XbCoord	XBCoord_CNSTMOD	X coordinate for conductor (vertical coordinates – feet) on position2.
Construction – Yb Coordinate	YbCoord	YBCoord_CNSTMOD	Y coordinate for conductor (vertical coordinates – feet) on position2. Usually, position2 is the highest position, which means its Y coordinate is biggest among all positions.

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Construction – Xc Coordinate	XcCoord	XCCOORD_CNSTMOD	X coordinate for conductor (vertical coordinates – feet) on position3. Usually, position3 is the rightmost position, which means its X coordinate is the biggest among all positions.
Construction – Yc Coordinate	YcCoord	YCCOORD_CNSTMOD	Y coordinate for conductor (vertical coordinates – feet) on position3.
Construction – Xn Coordinate	XnCoord	XNCOORD_CNSTMOD	X coordinate for conductor (vertical coordinates – feet) on positionN. Usually, this positionN is for neutral conductor.
Construction – Yn Coordinate	YnCoord	YNCOORD_CNSTMOD	Y coordinate for conductor (vertical coordinates – feet) on positionN.

4.4. Series Reactor Model

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Name	Name	NAME_SERIESMOD	Series reactor model name.
Series Reactor – R	R	R_SERIESMOD	Nominal resistance (ohms).
Series Reactor – X	X	X_SERIESMOD	Nominal reactance (ohms).
Oplimits(1): Name	CIMDnom.ConductorOpLimit. Name	OPLIMITNAMEn_SERIESMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(1): Unit	CIMDnom.ConductorOpLimit. Unit	OPLIMITUNITn_SERIESMOD	Enumerated limit units, 1 = Primary amps, 2 = KVA, 3 = MVA.
Oplimits(1): Limit	CIMDnom.ConductorOpLimit.Limit	OPLIMITn_SERIESMOD	Value of the operational limit.
Oplimits(2): Name	CIMDnom.ConductorOpLimit. Name	OPLIMITNAMEn_SERIESMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(2): Unit	CIMDnom.ConductorOpLimit. Unit	OPLIMITUNITn_SERIESMOD	Enumerated limit units, 1 = Primary amps, 2 = KVA, 3 = MVA.
Oplimits(2): Limit	CIMDnom.ConductorOpLimit.Limit	OPLIMITn_SERIESMOD	Value of the operational limit.
Oplimits(3): Name	CIMDnom.ConductorOpLimit. Name	OPLIMITNAMEn_SERIESMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Oplimits(3): Unit	CIMDnom.ConductorOpLimit.Unit	OPLIMITUNITn_SERIESMOD	Enumerated limit units, 1 = Primary amps, 2 = KVA, 3 = MVA.
Oplimits(3): Limit	CIMDnom.ConductorOpLimit.Limit	OPLIMITn_SERIESMOD	Value of the operational limit.
Oplimits(4): Name	CIMDnom.ConductorOpLimit.Name	OPLIMITNAMEn_SERIESMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(4): Unit	CIMDnom.ConductorOpLimit.Unit	OPLIMITUNITn_SERIESMOD	Enumerated limit units, 1 = Primary amps, 2 = KVA, 3 = MVA.
Oplimits(4): Limit	CIMDnom.ConductorOpLimit.Limit	OPLIMITn_SERIESMOD	Value of the operational limit.
Oplimits(5): Name	CIMDnom.ConductorOpLimit.Name	OPLIMITNAMEn_SERIESMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(5): Unit	CIMDnom.ConductorOpLimit.Unit	OPLIMITUNITn_SERIESMOD	Enumerated limit units, 1 = Primary amps, 2 = KVA, 3 = MVA.
Oplimits(5): Limit	CIMDnom.ConductorOpLimit.Limit	OPLIMITn_SERIESMOD	Value of the operational limit.

4.5. Shunt Impedance Model

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Name	Name	NAME_SHIMPMOD	Shunt impedance model name.
Shunt Impedance – P Total	Ptotal	PTOTAL_SMIMPMOD	Real power loss for a capacitor bank (kW).
Shunt Impedance – Switch Cost	Switchcost	SWITCOST_SHIMPMOD	Default cost of performing a switching operation (\$/operation).
Shunt Impedance – Close Switch	CloseSwitch	CLOSESW_SHIMPMOD	Limit on switching electrically close capacitors (ohms separation).
Shunt Impedance – Ind. Phase Op.	IndPhaseOp	INDPHASEOP_SHIMPMOD	Independent phase switching flag (true = device can be switched phase independently, false = ganged operation).

4.6. Switch Model

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Name	Name	NAME_SWMOD	Switch model name.
Switch – Type	SwType	SWTYPE_SWMOD	Enumerated switch type (to be determined).
Switch – Load Brk	LoadBrk	LOADBRK_SWMOD	Flag indicating switch has load break capability (true = has load break capability).
Switch – Cost	SwitCost	SWITCOST_SWMOD	Default cost of performing a switching operation (units).
Switch – Ganged?	Ganged	GANGED_SWMOD	Flag indicating ganged switch operation.
Switch – Reversible?	Reversible	REVERSEREC_SWMOD	Flag indicating reversible recloser.
Switch – Unique Name?	UniqueName	UNIQUENAME_SWMOD	Flag that indicates that the device has a unique name.
Switch – Telemetered?	Telemetered	TELEMETERED_SWMOD	Flag that indicates that the device is telemetered (true = telemetered).
Switch – Controllable?	Controllable	CONTROLLABLE_SWMOD	Flag that indicates if the device is controllable (true = controllable).
Switch – Bitmap Scheme	BitmapScheme	BITMAPSCHEME_SWMOD	Name of the bitmap scheme (default is blank).
Switch – Size	Size	SIZE_SWMOD	Device size (most likely an amp rating).
Switch – Load Break Rating	LoadBreakRating	LOADBRKR_SWMOD	Load breaking rating (most likely an amp rating).
Oplimits(1): Name	CIMDnom.ConductorOpLimit.Name	OPLIMITNAMEn_SWMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(1): Unit	CIMDnom.ConductorOpLimit.Unit	OPLIMITUNITn_SWMOD	Enumerated limit units, 1 = Primary amps, 2 = KVA, 3 = MVA.
Oplimits(1): Limit	CIMDnom.ConductorOpLimit.Limit	OPLIMITn_SWMOD	Value of the operational limit.
Oplimits(2): Name	CIMDnom.ConductorOpLimit.Name	OPLIMITNAMEn_SWMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(2): Unit	CIMDnom.ConductorOpLimit.Unit	OPLIMITUNITn_SWMOD	Enumerated limit units, 1 = Primary amps, 2 = KVA, 3 = MVA.
Oplimits(2): Limit	CIMDnom.ConductorOpLimit.Limit	OPLIMITn_SWMOD	Value of the operational limit.

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Oplimits(3): Name	CIMDnom.ConductorOpLimit.Name	OPLIMITNAME_SWMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(3): Unit	CIMDnom.ConductorOpLimit.Unit	OPLIMITUNIT_SWMOD	Enumerated limit units, 1 = Primary amps, 2 = KVA, 3 = MVA.
Oplimits(3): Limit	CIMDnom.ConductorOpLimit.Limit	OPLIMIT_SWMOD	Value of the operational limit.
Oplimits(4): Name	CIMDnom.ConductorOpLimit.Name	OPLIMITNAME_SWMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(4): Unit	CIMDnom.ConductorOpLimit.Unit	OPLIMITUNIT_SWMOD	Enumerated limit units, 1 = Primary amps, 2 = KVA, 3 = MVA.
Oplimits(4): Limit	CIMDnom.ConductorOpLimit.Limit	OPLIMIT_SWMOD	Value of the operational limit.
Oplimits(5): Name	CIMDnom.ConductorOpLimit.Name	OPLIMITNAME_SWMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(5): Unit	CIMDnom.ConductorOpLimit.Unit	OPLIMITUNIT_SWMOD	Enumerated limit units, 1 = Primary amps, 2 = KVA, 3 = MVA.
Oplimits(5): Limit	CIMDnom.ConductorOpLimit.Limit	OPLIMIT_SWMOD	Value of the operational limit.

4.7. Tap Model

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Name	Name	NAME_TAPMOD	Tap changer model name.
Tap – Max Tap	MaxTap	MAXTAP_TAPMOD	Maximum tap step (maybe +16).
Tap – Min Tap	MinTap	MINTAP_TAPMOD	Minimum tap step (maybe -16).
Tap – Incr.	TapIncr	TAPINCR_TAPMOD	Tap increment (percent V).
Tap – Nominal	Nominal	NOMINAL_TAPMOD	Nominal tap setting (maybe 0).
Tap – Delay 1	TimeDelay1	TIMEDELAY1_TAPMODEL	Initial time delay (seconds).
Tap – Delay 2	TimeDelay2	TIMEDELAY2_TAPMODEL	Subsequent time delay (seconds).
Tap – Cost	Tapcost	TAPCOST_TAPMODEL	Tap changer relative cost (units).

4.8. Transformer Model

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Name	Name	NAME_TFMOD	Transformer model name.
Transformer – Primary R1	PrimR1	PRIMR1_TFMOD	Primary positive sequence resistance (per unit on transformer power base).
Transformer – Primary X1	PrimX1	PRIMX1_TFMOD	Primary positive sequence reactance (per unit on transformer power base).
Transformer – Secondary1 R1	Sec1R1	SEC1R1_TFMOD	First secondary positive sequence resistance (per unit on transformer power base).
Transformer – Secondary1 X1	Sec1X1	SEC1X1_TFMOD	First secondary positive sequence reactance (per unit on transformer power base).
Transformer – Secondary2 R1	Sec2R1	SEC2R1_TFMOD	Second secondary positive sequence resistance (per unit on transformer power base).
Transformer – Secondary2 X1	Sec2X1	SEC2X1_TFMOD	Second secondary positive sequence resistance (per unit on transformer power base).
Transformer – Coreloss	Coreloss	CORELOSS_TFMOD	Transformer core loss (kW).
Transformer – MagPf	MagPf	MAGPF_TFMOD	Magnetization power factor (optional if percent excitation current is provided).
Transformer – MagB0	MagB0	MAGB0_TFMOD	Zero sequence magnetization susceptance (optional).
Transformer – MagG0	MagG0	MAGG0_TFMOD	Zero sequence magnetization conductance (optional).
Transformer – Excite Current	ExciteCurrent	EXCITECURRENT_TFMOD	Average percent excitation current (optional if the magnetizing power factor is provided).
Transformer – Type	Type	TYPE_TFMOD	Enumerated transformer type. The following values are used: 1 = Secondary transformer 2 = Voltage regulator 3 = Power transformer 4 = Auto-transformer 5 = Step Up/Down transformer.
Transformer – Reg. Units	RegUnits	REGUNITS_TFMODEL	Enumerated flag indicating the voltage units that should be used for transformers with voltage regulation capability. The following values can be used: 1 = Per unit voltage 2 = Actual voltage (on a configurable voltage base).

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Transformer – TF Type (Integer)	TfType	TFTYPE_TFMODEL	Enumerated transformer type for linking the transformer model to a transformer type that is needed by the GE ADMS power flow.
Transformer – TF Type (String)	strTfType		Transformer type in string.
Transformer – Units	Units	UNITS_TFMODEL	Transformer rating display units – 1 = kVA, 2 = MVA.
Transformer – S Rating	SRating	SRATING_TFMODEL	Transformer rating (in kVA or MVA).
Transformer – S Rating (Tertiary Winding)	SRatingTer	SRATINGTERT_TFMOD	Transformer rating for tertiary winding (in kVA or MVA).
Transformer – Bitmap Scheme	BitMapScheme	BITMAPSCHEME_TFMODEL	Name of the bitmap scheme (default is blank).
Oplimits(1): Name	CIMDnom.Operating Limit.Name	OPLIMITNAMEn_TFMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(1): Unit	CIMDnom.Operating Limit.Unit	OPLIMITUNITn_TFMOD	Enumerated limit units, 1 = Primary amps, 2 = KVA, 3 = MVA.
Oplimits(1): Limit	CIMDnom.Operating Limit.Limit	OPLIMITn_TFMOD	Value of the operational limit.
Oplimits(1): Season	CIMDnom.Operating Limit.Season	OPLIMITSEASONn_TFMOD	The name of an associated season when the limit is applicable.
Oplimits(1): LimitEvent	CIMDnom.Operating Limit.LimitEvent	OPLIMITEVENTn_TFMOD	Flag indicating a SCADA alarm is issued by the Distribution Power Flow when there is a violation of this limit (True = issue alarm).
Oplimits(1): TempSched	CIMDnom.Operating Limit.TempSched	OPLIMITSCHEDn_TFMOD	The name of an associated temperature-based limit schedule.
Oplimits(2): Name	CIMDnom.Operating Limit.Name	OPLIMITNAMEn_TFMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(2): Unit	CIMDnom.Operating Limit.Unit	OPLIMITUNITn_TFMOD	Enumerated limit units, 1 = Primary amps, 2 = KVA, 3 = MVA.
Oplimits(2): Limit	CIMDnom.Operating Limit.Limit	OPLIMITn_TFMOD	Value of the operational limit.
Oplimits(2): Season	CIMDnom.Operating Limit.Season	OPLIMITSEASONn_TFMOD	The name of an associated season when the limit is applicable.
Oplimits(2): LimitEvent	CIMDnom.Operating Limit.LimitEvent	OPLIMITEVENTn_TFMOD	Flag indicating a SCADA alarm is issued by the Distribution Power Flow when there is a violation of this limit (True = issue alarm).
Oplimits(2): TempSched	CIMDnom.Operating Limit.TempSched	OPLIMITSCHEDn_TFMOD	The name of an associated temperature-based limit schedule.

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Oplimits(3): Name	CIMDnom.Operating Limit.Name	OPLIMITNAME _n _TFMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(3): Unit	CIMDnom.Operating Limit.Unit	OPLIMITUNIT _n _TFMOD	Enumerated limit units, 1 = Primary amps, 2 = KVA, 3 = MVA.
Oplimits(3): Limit	CIMDnom.Operating Limit.Limit	OPLIMIT _n _TFMOD	Value of the operational limit.
Oplimits(3): Season	CIMDnom.Operating Limit.Season	OPLIMITSEASON _n _TFMOD	The name of an associated season when the limit is applicable.
Oplimits(3): LimitEvent	CIMDnom.Operating Limit.LimitEvent	OPLIMITEVENT _n _TFMOD	Flag indicating a SCADA alarm is issued by the Distribution Power Flow when there is a violation of this limit (True = issue alarm).
Oplimits(3): TempSched	CIMDnom.Operating Limit.TempSched	OPLIMITSCH _n _TFMOD	The name of an associated temperature-based limit schedule.
Oplimits(4): Name	CIMDnom.Operating Limit.Name	OPLIMITNAME _n _TFMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(4): Unit	CIMDnom.Operating Limit.Unit	OPLIMITUNIT _n _TFMOD	Enumerated limit units, 1 = Primary amps, 2 = KVA, 3 = MVA.
Oplimits(4): Limit	CIMDnom.Operating Limit.Limit	OPLIMIT _n _TFMOD	Value of the operational limit.
Oplimits(4): Season	CIMDnom.Operating Limit.Season	OPLIMITSEASON _n _TFMOD	The name of an associated season when the limit is applicable.
Oplimits(4): LimitEvent	CIMDnom.Operating Limit.LimitEvent	OPLIMITEVENT _n _TFMOD	Flag indicating a SCADA alarm is issued by the Distribution Power Flow when there is a violation of this limit (True = issue alarm).
Oplimits(4): TempSched	CIMDnom.Operating Limit.TempSched	OPLIMITSCH _n _TFMOD	The name of an associated temperature-based limit schedule.
Oplimits(5): Name	CIMDnom.Operating Limit.Name	OPLIMITNAME _n _TFMOD	The name of the limit (such as Long Term, Medium Term, or Short Term).
Oplimits(5): Unit	CIMDnom.Operating Limit.Unit	OPLIMITUNIT _n _TFMOD	Enumerated limit units, 1 = Primary amps, 2 = KVA, 3 = MVA.
Oplimits(5): Limit	CIMDnom.Operating Limit.Limit	OPLIMIT _n _TFMOD	Value of the operational limit.
Oplimits(5): Season	CIMDnom.Operating Limit.Season	OPLIMITSEASON _n _TFMOD	The name of an associated season when the limit is applicable.
Oplimits(5): LimitEvent	CIMDnom.Operating Limit.LimitEvent	OPLIMITEVENT _n _TFMOD	Flag indicating a SCADA alarm is issued by the Distribution Power Flow when there is a violation of this limit (True = issue alarm).
Oplimits(5): TempSched	CIMDnom.Operating Limit.TempSched	OPLIMITSCH _n _TFMOD	The name of an associated temperature-based limit schedule.

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
OplimitsSec(1): Name	CIMDnom.Operating Limit.Name	OPLIMITNAMESECn_TFMODE	The name of the limit for secondary winding (such as Long Term, Medium Term, or Short Term).
OplimitsSec(1): Unit	CIMDnom.Operating Limit.Unit	OPLIMITUNITSECn_TFMODE	Enumerated limit units limit for secondary winding, 1 = Primary amps, 2 = KVA, 3 = MVA.
OplimitsSec(1): Limit	CIMDnom.Operating Limit.Limit	OPLIMITUNITSECn_TFMODEx	Value of the operational limit for secondary winding.
OplimitsSec(1): Season	CIMDnom.Operating Limit.Season	OPLIMITSEASONSECn_TFMODE	The name of an associated season when the limit is applicable for secondary winding.
OplimitsSec(1): LimitEvent	CIMDnom.Operating Limit.LimitEvent	OPLIMITEVENTSECn_TFMODE	Flag indicating a SCADA alarm is issued by the Distribution Power Flow when there is a violation of this limit (True = issue alarm).
OplimitsSec(1): TempSched	CIMDnom.Operating Limit.TempSched	OPLIMITSCHEDSECn_TFMODE	The name of an associated temperature-based limit schedule for secondary winding.
OplimitsSec (2): Name	CIMDnom.Operating Limit.Name	OPLIMITNAMESECn_TFMODE	The name of the limit for secondary winding (such as Long Term, Medium Term, or Short Term).
OplimitsSec (2): Unit	CIMDnom.Operating Limit.Unit	OPLIMITUNITSECn_TFMODE	Enumerated limit units limit for secondary winding, 1 = Primary amps, 2 = KVA, 3 = MVA.
OplimitsSec (2): Limit	CIMDnom.Operating Limit.Limit	OPLIMITSECn_TFMODE	Value of the operational limit for secondary winding.
OplimitsSec(2): Season	CIMDnom.Operating Limit.Season	OPLIMITSEASONSECn_TFMODE	The name of an associated season when the limit is applicable for secondary winding.
OplimitsSec(2): LimitEvent	CIMDnom.Operating Limit.LimitEvent	OPLIMITEVENTSECn_TFMODE	Flag indicating a SCADA alarm is issued by the Distribution Power Flow when there is a violation of this limit (True = issue alarm).
OplimitsSec(2): TempSched	CIMDnom.Operating Limit.TempSched	OPLIMITSCHEDSECn_TFMODE	The name of an associated temperature-based limit schedule for secondary winding.
OplimitsSec (3): Name	CIMDnom.Operating Limit.Name	OPLIMITNAMESECn_TFMODE	The name of the limit for secondary winding (such as Long Term, Medium Term, or Short Term).
OplimitsSec (3): Unit	CIMDnom.Operating Limit.Unit	OPLIMITUNITSECn_TFMODE	Enumerated limit units limit for secondary winding, 1 = Primary amps, 2 = KVA, 3 = MVA.
OplimitsSec (3): Limit	CIMDnom.Operating Limit.Limit	OPLIMITSECn_TFMODE	Value of the operational limit for secondary winding.

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
OplimitsSec(3): Season	CIMDnom.Operating Limit.Season	OPLIMITSEASONSECn_TFMODE	The name of an associated season when the limit is applicable for secondary winding.
OplimitsSec(3): LimitEvent	CIMDnom.Operating Limit.LimitEvent	OPLIMITEVENTSECn_TFMODE	Flag indicating a SCADA alarm is issued by the Distribution Power Flow when there is a violation of this limit (True = issue alarm).
OplimitsSec(3): TempSched	CIMDnom.Operating Limit.TempSched	OPLIMITSCHEDSECn_TFMODE	The name of an associated temperature-based limit schedule for secondary winding.
OplimitsSec (4): Name	CIMDnom.Operating Limit.Name	OPLIMITNAMESECn_TFMODE	The name of the limit for secondary winding (such as Long Term, Medium Term, or Short Term).
OplimitsSec (4): Unit	CIMDnom.Operating Limit.Unit	OPLIMITUNITSECn_TFMODE	Enumerated limit units limit for secondary winding, 1 = Primary amps, 2 = KVA, 3 = MVA.
OplimitsSec (4): Limit	CIMDnom.Operating Limit.Limit	OPLIMITSECn_TFMODE	Value of the operational limit for secondary winding.
OplimitsSec(4): Season	CIMDnom.Operating Limit.Season	OPLIMITSEASONSECn_TFMODE	The name of an associated season when the limit is applicable for secondary winding.
OplimitsSec(4): LimitEvent	CIMDnom.Operating Limit.LimitEvent	OPLIMITEVENTSECn_TFMODE	Flag indicating a SCADA alarm is issued by the Distribution Power Flow when there is a violation of this limit (True = issue alarm).
OplimitsSec(4): TempSched	CIMDnom.Operating Limit.TempSched	OPLIMITSCHEDSECn_TFMODE	The name of an associated temperature-based limit schedule for secondary winding.
OplimitsSec (5): Name	CIMDnom.Operating Limit.Name	OPLIMITNAMESECn_TFMODE	The name of the limit for secondary winding (such as Long Term, Medium Term, or Short Term).
OplimitsSec (5): Unit	CIMDnom.Operating Limit.Unit	OPLIMITUNITSECn_TFMODE	Enumerated limit units limit for secondary winding, 1 = Primary amps, 2 = KVA, 3 = MVA.
OplimitsSec (5): Limit	CIMDnom.Operating Limit.Limit	OPLIMITSECn_TFMODE	Value of the operational limit for secondary winding.
OplimitsSec(5): Season	CIMDnom.Operating Limit.Season	OPLIMITSEASONSECn_TFMODE	The name of an associated season when the limit is applicable for secondary winding.
OplimitsSec(5): LimitEvent	CIMDnom.Operating Limit.LimitEvent	OPLIMITEVENTSECn_TFMODE	Flag indicating a SCADA alarm is issued by the Distribution Power Flow when there is a violation of this limit (True = issue alarm).
OplimitsSec(5): TempSched	CIMDnom.Operating Limit.TempSched	OPLIMITSCHEDSECn_TFMODE	The name of an associated temperature-based limit schedule for secondary winding.

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
OplimitsTer(1): Name	CIMDnom.Operating Limit.Name	OPLIMITNAMETERn_TFMOD	The name of the limit for tertiary winding (such as Long Term, Medium Term, or Short Term).
OplimitsTer (1): Unit	CIMDnom.Operating Limit.Unit	OPLIMITUNITTERn_TFMOD	Enumerated limit units limit for tertiary winding, 1 = Primary amps, 2 = KVA, 3 = MVA.
OplimitsTer (1): Limit	CIMDnom.Operating Limit.Limit	OPLIMITTERn_TFMOD	Value of the operational limit for tertiary winding.
OplimitsTer(1): Season	CIMDnom.Operating Limit.Season	OPLIMITSEASONTERn_TFMOD	The name of an associated season when the limit is applicable for tertiary winding.
OplimitsTer(1): LimitEvent	CIMDnom.Operating Limit.LimitEvent	OPLIMITEVENTTERn_TFMOD	Flag indicating a SCADA alarm is issued by the Distribution Power Flow when there is a violation of this limit (True = issue alarm).
OplimitsTer(1): TempSched	CIMDnom.Operating Limit.TempSched	OPLIMITSCHEDTERn_TFMOD	The name of an associated temperature-based limit schedule for tertiary winding.
OplimitsTer (2): Name	CIMDnom.Operating Limit.Name	OPLIMITNAMETERn_TFMOD	The name of the limit for tertiary winding (such as Long Term, Medium Term, or Short Term).
OplimitsTer (2): Unit	CIMDnom.Operating Limit.Unit	OPLIMITUNITTERn_TFMOD	Enumerated limit units limit for tertiary winding, 1 = Primary amps, 2 = KVA, 3 = MVA.
OplimitsTer (2): Limit	CIMDnom.Operating Limit.Limit	OPLIMITTERn_TFMOD	Value of the operational limit for tertiary winding.
OplimitsTer(2): Season	CIMDnom.Operating Limit.Season	OPLIMITSEASONTERn_TFMOD	The name of an associated season when the limit is applicable for tertiary winding.
OplimitsTer(2): LimitEvent	CIMDnom.Operating Limit.LimitEvent	OPLIMITEVENTTERn_TFMOD	Flag indicating a SCADA alarm is issued by the Distribution Power Flow when there is a violation of this limit (True = issue alarm).
OplimitsTer(2): TempSched	CIMDnom.Operating Limit.TempSched	OPLIMITSCHEDTERn_TFMOD	The name of an associated temperature-based limit schedule for tertiary winding.
OplimitsTer (3): Name	CIMDnom.Operating Limit.Name	OPLIMITNAMETERn_TFMOD	The name of the limit for tertiary winding (such as Long Term, Medium Term, or Short Term).
OplimitsTer (3): Unit	CIMDnom.Operating Limit.Unit	OPLIMITUNITTERn_TFMOD	Enumerated limit units limit for tertiary winding, 1 = Primary amps, 2 = KVA, 3 = MVA.
OplimitsTer (3): Limit	CIMDnom.Operating Limit.Limit	OPLIMITTERn_TFMOD	Value of the operational limit for tertiary winding.

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
OplimitsTer(3): Season	CIMDnom.Operating Limit.Season	OPLIMITSEASONTERn_TFMODE	The name of an associated season when the limit is applicable for tertiary winding.
OplimitsTer(3): LimitEvent	CIMDnom.Operating Limit.LimitEvent	OPLIMITEVENTTERn_TFMODE	Flag indicating a SCADA alarm is issued by the Distribution Power Flow when there is a violation of this limit (True = issue alarm).
OplimitsTer(3): TempSched	CIMDnom.Operating Limit.TempSched	OPLIMITSCHEDTERn_TFMODE	The name of an associated temperature-based limit schedule for tertiary winding.
OplimitsTer (4): Name	CIMDnom.Operating Limit.Name	OPLIMITNAMETERn_TFMODE	The name of the limit for tertiary winding (such as Long Term, Medium Term, or Short Term).
OplimitsTer (4): Unit	CIMDnom.Operating Limit.Unit	OPLIMITUNITTERn_TFMODE	Enumerated limit units limit for tertiary winding, 1 = Primary amps, 2 = KVA, 3 = MVA.
OplimitsTer (4): Limit	CIMDnom.Operating Limit.Limit	OPLIMITTERn_TFMODE	Value of the operational limit for tertiary winding.
OplimitsTer(4): Season	CIMDnom.Operating Limit.Season	OPLIMITSEASONTERn_TFMODE	The name of an associated season when the limit is applicable for tertiary winding.
OplimitsTer(4): LimitEvent	CIMDnom.Operating Limit.LimitEvent	OPLIMITEVENTTERn_TFMODE	Flag indicating a SCADA alarm is issued by the Distribution Power Flow when there is a violation of this limit (True = issue alarm).
OplimitsTer(4): TempSched	CIMDnom.Operating Limit.TempSched	OPLIMITSCHEDTERn_TFMODE	The name of an associated temperature-based limit schedule for tertiary winding.
OplimitsTer (5): Name	CIMDnom.Operating Limit.Name	OPLIMITNAMETERn_TFMODE	The name of the limit for tertiary winding (such as Long Term, Medium Term, or Short Term).
OplimitsTer (5): Unit	CIMDnom.Operating Limit.Unit	OPLIMITUNITTERn_TFMODE	Enumerated limit units limit for tertiary winding, 1 = Primary amps, 2 = KVA, 3 = MVA.
OplimitsTer (5): Limit	CIMDnom.Operating Limit.Limit	OPLIMITTERn_TFMODE	Value of the operational limit for tertiary winding.
OplimitsTer(5): Season	CIMDnom.Operating Limit.Season	OPLIMITSEASONTERn_TFMODE	The name of an associated season when the limit is applicable for tertiary winding.
OplimitsTer(5): LimitEvent	CIMDnom.Operating Limit.LimitEvent	OPLIMITEVENTTERn_TFMODE	Flag indicating a SCADA alarm is issued by the Distribution Power Flow when there is a violation of this limit (True = issue alarm).
OplimitsTer(5): TempSched	CIMDnom.Operating Limit.TempSched	OPLIMITSCHEDTERn_TFMODE	The name of an associated temperature-based limit schedule for tertiary winding.

4.9. Voltage Level Model

Substation Editor Column Name	Global Internals XML Field Name	Corresponding Global CSV Field Name	Description
Name	Name	NAME_VOLTLEVEL	Voltage level model name.
Volt Level – Value [kV]	ValuekV	VALUEKV_VOLTLEVEL	Voltage level value (in kV L-L).
Volt Level – Dist. V.L.	DistVL	DISTVL_VOLTLEVEL	Flag indicating voltage level is a distribution or transmission voltage level. (True = distribution, false = transmission).
Volt Level – Color	Color	COLOR_VOLTLEVEL	Color -- for UI purposes.

5. SSE Tools

SSE Tools are used for batch processing common modeling tasks (refer to the *Substation Editor User Guide* for specifics). They are installed in the \Tools\bin directory and are expected to run in the same environment as the Substation Editor (although they may use different directories).

5.1. About Log Filtering

For a discussion of differences in logging between Substation Editor and the SSE Tools, refer to the *Substation Editor User Guide*. For those tools which accept a -CONFIG argument, the messages logged to the console can be filtered by severity according to the SubstationEditorConfiguration <LogLevel> property. The correlation between the LoggingLevels names and the configuration value is:

Severity (LoggingLevels)	<LogLevel>
Fatal	0
Error	1
Warning	2
Info	3
Debug1	4
Debug2	5
Debug3	6
Debug4	7
Debug5	8
Debug6	9
Debug7	10
Debug8	11
Debug9	12
Debug10	13

Note that Fatal messages cannot be filtered out.

5.2. Dependency on Substation Editor

Before building any of the SSE Tools, Substation Editor must have been successfully built on the same platform and in the same configuration as the SSE Tools.

The supported (\$Configuration)|(\$Platform) pairs are:

- Release | x64
- Debug | x64

Like Substation Editor, the SSE Tools are only supported on Windows 7 (although they might work on Windows Server platforms).

At runtime, the SSE Tools expect to find all of the Substation Editor assemblies in the same directory as the SSE Tool executable (for example, in the \Eterra\Distribution\Tools\bin directory).

5.3. SSETools_Common

This is a library of supporting functions for the SSE Tools. The following assemblies are needed in the same directory as the SSE Tool executable:

- SSETools_Common.dll
- SSETools_Common.resources.dll

It is helpful to have the .pdb and .config files in the same directory.

5.4. BatchPostageStamps

The ability to generate Postage Stamp Displays and their measurements in batch mode is provided by two executables: `BatchPostageStamps.exe` and `PrepBPSIncludeList.exe`.

5.4.1. BatchPostageStamps.exe

BatchPostageStamps can be run outside of the ADMS installation root directory, but it needs the Substation Editor and SSETools_Common assemblies to be present in the same directory.

All of the BatchPostageStamps assemblies need to be in the same directory:

- BatchPostageStamps.exe
- BatchPostageStamps.resources.dll

It is helpful to have the .pdb and .config files in the same directory.

5.4.2. PrepBPSIncludeList.exe

PrepBPSIncludeList can be run outside of the ADMS installation root directory, but it needs the Substation Editor and SSETools_Common assemblies to be present in the same directory.

All of the PrepBPSIncludeList assemblies need to be in the same directory:

- PrepBPSIncludeList.exe
- PrepBPSIncludeList.resources.dll

It is helpful to have the .pdb and .config files in the same directory.

5.4.3. PurgePlacements.exe

PurgePlacements can be run outside of the ADMS installation root directory, but it needs the Substation Editor and SSETools_Common assemblies to be present in the same directory.

All of the PurgePlacements assemblies need to be in the same directory:

- PurgePlacements.exe
- PurgePlacements.resources.dll

It is helpful to have the .pdb and .config files in the same directory.

5.5. BulkEdit

The ability to update electrical characteristics in batch mode is provided by two executables: BulkExtract.exe and BulkMerge.exe.

5.5.1. BulkEditCommon.exe

This is a library of supporting functions for the BulkEdit executables. The following assemblies are needed in the same directory as the BulkExtract and/or BulkMerge executable:

- BulkEditCommon.dll
- BulkEditCommon.resources.dll

It is helpful to have the .pdb and .config files in the same directory.

5.5.2. BulkExtract.exe

BulkExtract can be run outside of the ADMS installation root directory, but it needs the Substation Editor assemblies, the SSETools_Common assemblies, and the BulkEditCommon assemblies to be present in the same directory.

All of the BulkExtract assemblies need to be in the same directory:

- BulkExtract.exe
- BulkExtract.resources.dll

It is helpful to have the .pdb and .config files in the same directory.

5.5.3. BulkMerge.exe

BulkMerge can be run outside of the ADMS installation root directory, but it needs the Substation Editor assemblies, the SSETools_Common assemblies, and the BulkEditCommon assemblies to be

present in the same directory.

All of the BulkMerge assemblies need to be in the same directory:

- BulkMerge.exe
- BulkMerge.resources.dll

It is helpful to have the .pdb and .config files in the same directory.